

TAMIL NADU STATE BOARD - STANDARD 11 PHYSICS VOLUME 1

CHAPTER 3: LAWS OF MOTION

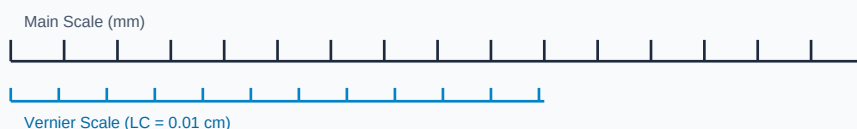
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Physics Volume 1
Chapter Name:	Laws of Motion

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division

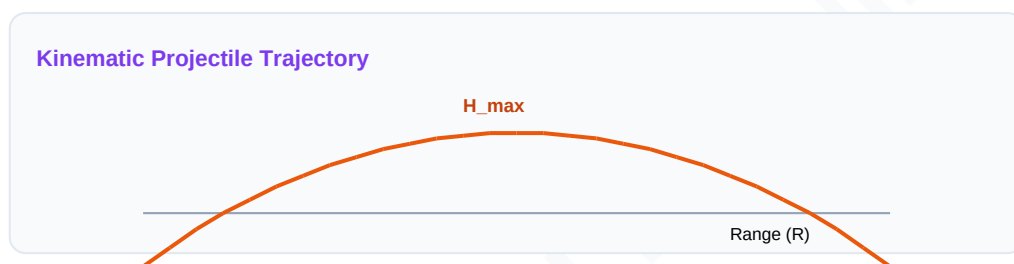


Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Inertia	The inherent property of a body to resist any change in its state of rest or uniform motion.
Impulse	A large force acting on a body for a short duration, measured as the product of force and time, equal to change in momentum.
Coefficient of Friction	The ratio of the frictional force to the normal force acting between two contact surfaces.

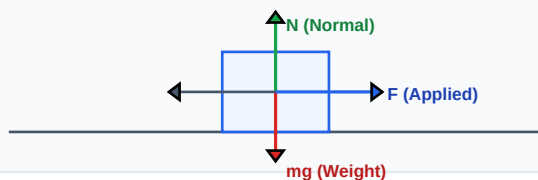
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Newton's Second Law: $F = dp/dt = m \cdot a$
- Impulse: $J = F \cdot \Delta t = \Delta p$
- Static Friction: $f_s \leq \mu_s \cdot N$ | Kinetic: $f_k = \mu_k \cdot N$
- Apparent Weight in Lift: $R = m(g \pm a)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

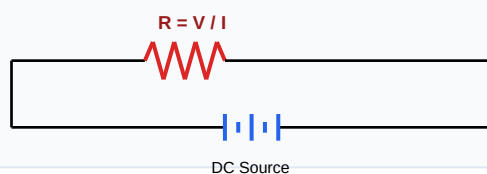
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A body of mass 5 kg is acted on by a force of 20 N. Find its acceleration.

Step-by-step Solution:

Apply $F = ma$: $a = F / m = 20 / 5 = 4 \text{ m/s}^2$.

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

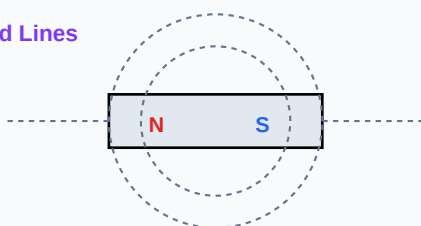
HOTS Challenge 1: Why is it easier to pull a lawn mower than to push it?

Analytical Solution Strategy:

When pushing: vertical component of force acts downwards, increasing normal force $N = mg + F \sin(\theta)$. Thus, friction $f = \mu * N$ increases.

When pulling: vertical component of force acts upwards, reducing normal force $N = mg - F \sin(\theta)$. Thus, friction decreases, making pulling easier.

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Newton's three laws of motion.

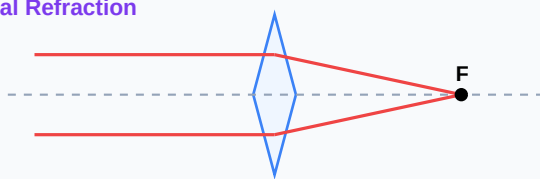
Q2 (2-Mark): Explain the law of conservation of linear momentum with an example.

Q3 (2-Mark): What is banking of roads? State its expression for speed limits.

Q4 (5-Mark): A 10kg mass is in a lift moving down with $a=2 \text{ m/s}^2$. Find tension in the cable.

Q5 (5-Mark): Distinguish between static friction and kinetic friction.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Automobile Engineering

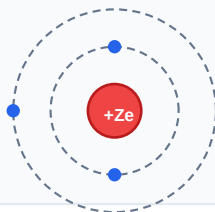
Crumple zones are designed to increase collision time, reducing impact force on passengers.

Application Case: Athletics

Running tracks are designed with specific friction boundaries to optimize spikes traction.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Sinusoidal AC Voltage Waveform

